Introduction to LaTeX

Session 4: Beamer and TikZ

$$P(y) = \int P(y|x)\, P(x)\, \mathrm{d}x$$

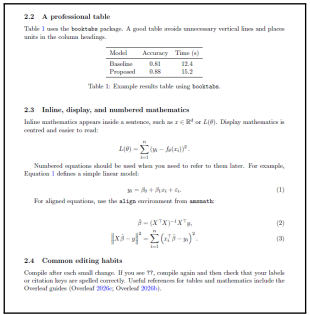
$$x_{n+1} = \lambda x_n (1 - x_n)$$

$$\lim_{x \to \infty} \frac{1}{1 + x^{-1}}$$

$$\sum_{n=1}^{N} n(n-1)$$

$$\int_b^a f(x)\, \mathrm{d}x$$

Recap: Session 3



2.2 A professional table

Table 1 uses the `booktabs` package. A good table avoids unnecessary vertical lines and places units in the column headings.

Model	Accuracy	Time (s)
Baseline	0.81	12.4
Proposed	0.88	15.2

Table 1: Example results table using `booktabs`.

2.3 Inline, display, and numbered mathematics

Inline mathematics appears inside a sentence, such as $x \in \mathbb{R}^d$ or $L(\theta)$. Display mathematics is centred and easier to read:

$$L(\theta) = \sum_{i=1}^{N} (y_i - f_\theta(x_i))^2 .$$

Numbered equations should be used when you need to refer to them later. For example, Equation 1 defines a simple linear model:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i. \tag{1}$$

For aligned equations, use the `align` environment from **amsmath**:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \tag{2}$$

$$\left\| X\hat{\beta} - y \right\|^2 = \sum_{i=1}^{N} \left(x_i^\top \hat{\beta} - y_i \right)^2 . \tag{3}$$

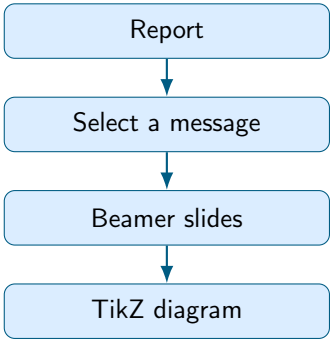
2.4 Common editing habits

Compile after each small change. If you see ??, compile again and then check that your labels or citation keys are spelled correctly. Useful references for tables and mathematics include the Overleaf guides (Overleaf 2026c, Overleaf 2026b).

- Tables: `tabular`, `table`, `booktabs`, captions and labels.
- Bibliography: `references.bib`, `biblatex`, citation commands and `\printbibliography`.
- Mathematics: inline maths, display maths, numbered equations, `amsmath`.
- Cross-references: the pattern `caption/label/ref` also applies to tables and equations.

Outline

- Why use Beamer?
- Beamer basics: frames, blocks, columns and overlays
- Short exercise: build a Beamer presentation
- TikZ basics: coordinates, nodes and arrows
- Short exercise: add a TikZ diagram
- Final checks before export



Beamer and TikZ

Before Beamer: slides are not compressed reports

A report can be dense

- paragraphs;
- full tables;
- detailed methods;
- complete derivations;
- long references.

A slide should guide attention

- one purpose;
- one visual anchor;
- minimal text;
- spoken explanation;
- concise citations.

A simple five-slide structure

Slide	Purpose	Good visual anchor
1. Title	Who are you and what is the project?	title and affiliation
2. Motivation	Why should the audience care?	motivating image or schematic
3. Method/data	What did you do?	flowchart or pipeline
4. Result/example	What did you find?	figure, table or key equation
5. Takeaway	What should the audience remember?	one-sentence conclusion

Rule

Slides are a guided route through the report, not a smaller-font version of the report.

1 Beamer

Why Beamer?

Good when you need

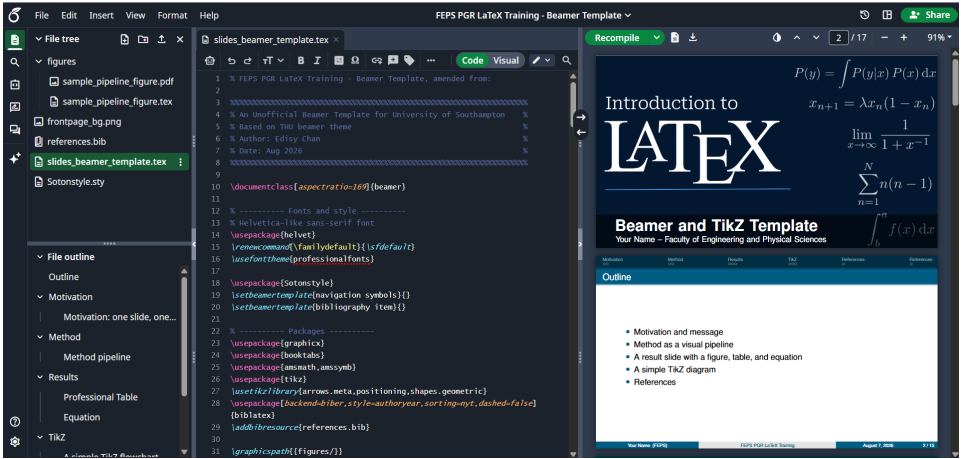
- mathematics that matches your report;
- reusable figures, tables and macros;
- consistent formatting across a whole deck;
- version-controlled source files.

Less ideal when you need

- highly free-form visual design;
- heavy image editing;
- drag-and-drop layouts.

Source: Overleaf Beamer guide (Overleaf 2026a); Beamer user guide (Tantau, Wright, and Miletic 2026).

What a Beamer project looks like

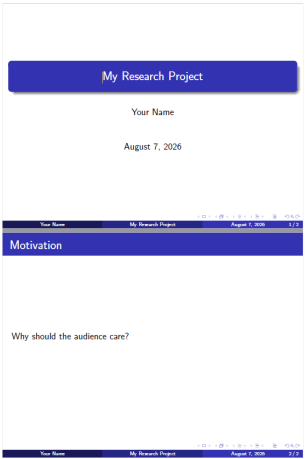


A minimal Beamer file

```
\documentclass{beamer}
\usetheme{Madrid}
\title{My Research Project}
\author{Your Name}
\date{\today}

\begin{document}
\begin{frame}
  \titlepage
\end{frame}

\begin{frame}{Motivation}
  Why should the audience care?
\end{frame}
\end{document}
```



Source: Overleaf Beamer guide (Overleaf 2026a); Oxford Beamer training (Hutchinson 2014).

What does the minimal Beamer file do?

Preamble

- `\documentclass{beamer}` creates a presentation rather than an article.
- (optional) `\usetheme{Madrid}` chooses a visual theme.
- `\title`, `\author`, and `\date` store information for the title slide.

Document body

- `\begin{document}` starts the presentation content.
- Each `frame` environment defines one presentation frame.
- `\titlepage` prints the stored title, author, and date.
- `\begin{frame}{Motivation}` creates a slide titled *Motivation*.

Key idea

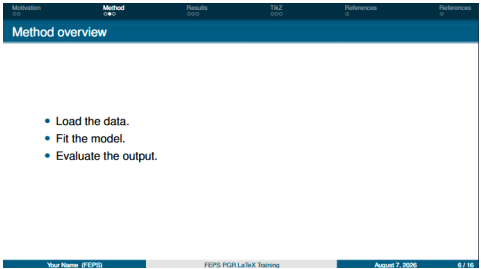
A Beamer document has the same basic structure as other LaTeX documents: a preamble followed by the document body, but the body is organised into `frame` environments.

Source: Overleaf Beamer guide (Overleaf 2026a).

The `frame` environment is the basic unit

```
\section{Method}

\begin{frame}{Method overview}
\begin{itemize}
  \item Load the data.
  \item Fit the model.
  \item Evaluate the output.
\end{itemize}
\end{frame}
```



Frame habits

- Give most slides a title.
- Use sections to organise the presentation.
- Keep each slide focused on one main message.
- Avoid putting a full paragraph on a slide.

Source: Oxford Beamer training (Hutchinson 2014); Beamer user guide (Tantau, Wright, and Miletic 2026).

Blocks, columns and lists

```
\begin{columns}
\column{0.48\textwidth}
\begin{block}{Claim}
Important message.
\end{block}
\column{0.48\textwidth}
\begin{itemize}
  \item Evidence 1
  \item Evidence 2
\end{itemize}
\end{columns}
```

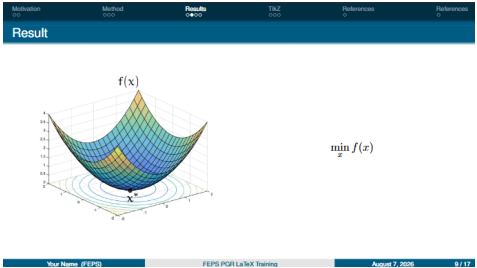
Claim

Important message.

- Evidence 1
- Evidence 2

Source: Overleaf Beamer guide (Overleaf 2026a); Oxford Beamer training (Hutchinson 2014).

```
\begin{frame}{Result}
\begin{columns}
\column{0.48\textwidth}
\includegraphics[width=\linewidth]{fig.png}
\column{0.48\textwidth}
\[
  \min_x f(x)
\]
\end{columns}
\end{frame}
```



Presentation rule

Not everything from the report belongs on a slide. Choose one figure, table or equation and explain its role clearly.

A Beamer theme controls

- headers, footers, colours, blocks and navigation;
- spacing and font sizes;
- the visual identity of the deck.

Practical advice

Choose one clean theme or template, then stop manually formatting every slide. Let the template carry the design.

Remark

These slides use a customised unofficial Soton Beamer theme adapted by Edisy Chan.

Source: Overleaf Beamer guide (Overleaf 2026a); Oxford Beamer training (Hutchinson 2014).

Overlays: useful, but optional

```
\begin{itemize}
  \item<1-> First point
  \item<2-> Second point
  \item<3-> Third point
\end{itemize}
```

- First point
- Second point
- Third point

<2-> means “show this item from overlay 2 onwards”.

Caution

Overlays can help pacing, but too many make source files harder to maintain.

Source: Beamer user guide (Tantau, Wright, and Miletic 2026); Stony Brook Beamer/TikZ training (Schnell 2017).

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Using your Session 3 report

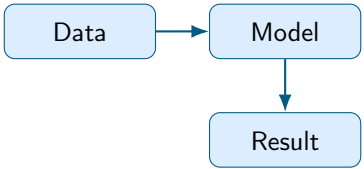
1. Create a new `.tex` file with `\documentclass{beamer}`.
2. Add the title, author and date, then create a title slide.
3. Add four content frames: motivation/background, method/data, result/example, and takeaway/next steps.
4. Reuse at least one figure, table or equation from your report.
5. Compile and check that each slide has one clear main message.

Checkpoint

You should now have a five-slide Beamer presentation that compiles correctly.

2 TikZ

- TikZ lets you draw diagrams directly in LaTeX source code.
- It is useful for flowcharts, block diagrams, annotated schematics and conceptual diagrams.
- It is not essential for all Beamer slides, but it is powerful when diagrams need to match the document style.



Goal

Know that TikZ exists, understand the basic syntax, and leave with one adaptable diagram.

Source: PGF/TikZ package (CTAN 2026); Stony Brook Beamer/TikZ training (Schnell 2017).

Before drawing: load TikZ

```
% preamble
\usepackage{tikz}
\usetikzlibrary{positioning, arrows.meta}
```

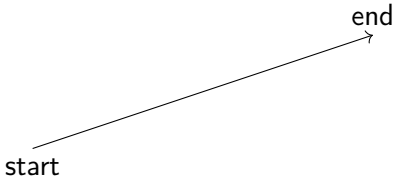
What these do

- `tikz` enables TikZ drawings.
- `positioning` enables relative positions such as `right=of`.
- `arrows.meta` provides additional arrow styles such as `Latex`.

Source: Overleaf TikZ guide (Overleaf 2026b); PGF/TikZ package (CTAN 2026).

TikZ basics: coordinates, paths and semicolons

```
\begin{tikzpicture}
  \draw[->] (0,0) -- (3,1);
  \node[below] at (0,0) {start};
  \node[above] at (3,1) {end};
\end{tikzpicture}
```



Key ideas

- Coordinates are written as `(x,y)`.
- `\draw` creates a path; `-` draws a straight line.
- Options such as `[->]`, `[above]`, and `[below]` control appearance and position.
- `\node` places text, and every TikZ command ends with a semicolon.

Source: Overleaf TikZ guide (Overleaf 2026b); Stony Brook Beamer/TikZ training (Schnell 2017).

Nodes, shapes and arrows

```
\begin{tikzpicture}[node distance=3.5cm]
\node[draw,rounded corners]
      (data) {Data};
\node[draw,rounded corners,
      right=of data]
      (model) {Model};
\draw[->] (data) -- (model);
\end{tikzpicture}
```



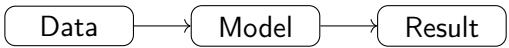
Key ideas

- Give nodes names, e.g. `(data)` and `(model)`.
- Options such as `draw` and `rounded corners` control the node appearance.
- Relative positioning such as `right=of data` places one node beside another.
- Named nodes can be connected directly with `\draw[->] (data) -- (model)`.

Source: Overleaf TikZ guide (Overleaf 2026b).

An adaptable flowchart

```
\begin{tikzpicture}[node distance=1cm]
\node[draw,rounded corners]
      (data) {Data};
\node[draw,rounded corners,
      right=of data]
      (model) {Model};
\node[draw,rounded corners,
      right=of model]
      (result) {Result};
\draw[->] (data) -- (model);
\draw[->] (model) -- (result);
\end{tikzpicture}
```



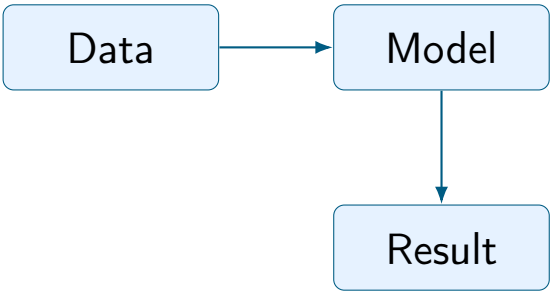
How to adapt

Change the node text first, then change the positions. Styling can come later.

Source: Adapted from Overleaf TikZ examples (Overleaf 2026b).

TikZ template: code

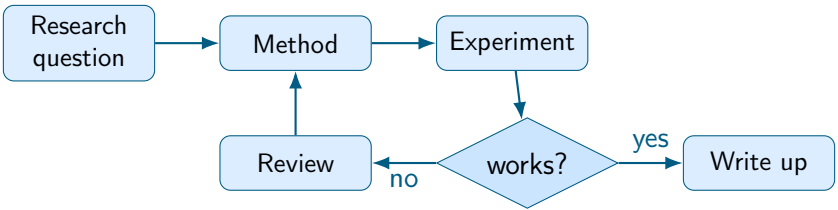
```
\begin{tikzpicture}[
  node distance=1.0cm,
  box/.style={draw=Soton,fill=Sotonb!50,rounded corners,
              minimum width=1.9cm,minimum height=0.75cm,align=center},
  arr/.style={-Latex,thick,Soton}
]
\node[box] (data) {Data};
\node[box,right=of data] (model) {Model};
\node[box,below=of model] (result) {Result};
\draw[arr] (data) -- (model);
\draw[arr] (model) -- (result);
\end{tikzpicture}
```



Reusable styles

`box` and `arr` are custom styles defined in the `tikzpicture`. They let us reuse the same formatting for several nodes and arrows.

A richer research workflow diagram



Point to note

For simple diagrams, TikZ source code can be clearer and more reproducible than drawing by hand in presentation software.

Source: TikZ examples for further adaptation (TeXample.net 2026).

Add one diagram to your Beamer deck

1. Start from the `Data -> Model -> Result` example.
2. Change the node text so that it represents part of your own research.
3. Change at least one node position, for example from `right=of` to `below=of`.
4. Adjust the arrows or layout if needed.
5. Add the diagram to an appropriate slide in your five-slide deck.

Checkpoint

The diagram should communicate part of your research, not simply reproduce the example.

3

Before you export

Checklist before export

- Does each slide have one main message?
- Are fonts readable from the back of a teaching room?
- Are figures cropped and large enough?
- Are equations necessary, explained and not too small?
- Do all citations, references and labels compile without warnings?
- Does the PDF still look correct after downloading from Overleaf?

Common Beamer and TikZ errors

Symptom	**First thing to check**
Content runs off the slide	Split the slide or reduce the amount of content.
TikZ error near a line	Check for a missing semicolon at the end of a TikZ command.
`right=of` or `below=of` fails	Load `\usetikzlibrary{positioning}`.
TikZ diagram is too large	Reduce `node distance`, node sizes, or adjust the scale.

4 Where to go next

Where to go next

- Use Overleaf's Beamer and TikZ guides for quick examples.
- Use the Beamer user guide when you need precise behaviour.
- Use TikZ examples when you want to adapt a diagram pattern.
- Keep your own small library of diagrams and slide components.

Source: Overleaf Beamer (Overleaf 2026a); Overleaf TikZ (Overleaf 2026b); Beamer user guide (Tantau, Wright, and Miletic 2026); TeXample (TeXample.net 2026).



Extra Slides

CTAN (2026). *PGF/TikZ – Create PostScript and PDF Graphics in TeX*. URL: https://ctan.org/pkg/pgf (visited on 07/29/2026).

Hutchinson, Susan (2014). *LaTeX Level 4: Creating Beamer Presentations*. URL: https://www.stats.ox.ac.uk/pub/susan/cdt/Creating_Presentations/Beamer_slides.pdf (visited on 07/29/2026).

Overleaf (2026a). *Beamer*. URL: https://www.overleaf.com/learn/latex/Beamer (visited on 07/29/2026).

Overleaf (2026b). *TikZ Package*. URL: https://www.overleaf.com/learn/latex/TikZ_package (visited on 07/29/2026).

Schnell, Christian (Apr. 7, 2017). *Introduction to Beamer and TikZ*. URL: https://www.math.stonybrook.edu/rtg/ProfessionalDevelopment/beamer-tikz.pdf (visited on 07/29/2026).

Tantau, Till, Joseph Wright, and Vedran Miletic (2026). *The Beamer Class User Guide*. URL: https://mirrors.ctan.org/macros/latex/contrib/beamer/doc/beameruserguide.pdf (visited on 07/29/2026).

TeXample.net (2026). *TikZ and PGF examples*. URL: https://texample.net/ (visited on 07/29/2026).

Thank You